M|P|W

Physics A

Chapters 1 & 2
Nick Papanikolaou

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **35**.
- The total number of marks may take into account some 'either/or' question choices.

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1. The table below shows the measurements recorded by a student for a solid metal sphere. The absolute uncertainties in the mass of the sphere and in its radius are also shown.

mass	100 ± 6 g
radius	1.60 ± 0.08 cm

What is the percentage uncertainty in the density of the sphere?

A. 1%
B. 11%
C. 16%
D. 21%

Your answer

[1]

2. A student determines the power *P* dissipated in a resistor. The measured values of the current **I** in the resistor and the resistance *R* of the resistor are:

$$I = (4.0 \pm 0.2)\ \text{A and } R = (3.0 \pm 0.3)\ \Omega$$

The equation $P = I^2R$ is used to calculate *P*.
What is the percentage uncertainty in the value of *P*?

A 15%
B 20%
C 25%
D 30%

Your answer **[1]**

3. Four students each carry out an experiment to determine the acceleration of free fall *g*.

Which is the **least** accurate value?

A $(9.0 \pm 1.0)\ \text{ms}^{-2}$
B $(9.5 \pm 0.1)\ \text{ms}^{-2}$
C $(9.6 \pm 0.4)\ \text{ms}^{-2}$
D $(9.7 \pm 0.2)\ \text{ms}^{-2}$

Your answer **[1]**

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4. Which is **not** an International System (S.I.) base unit?

A second (s)
B kelvin (K)
C kilogram (kg)
D coulomb (C)

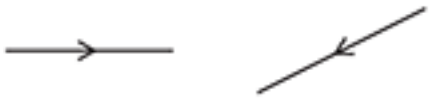
Your answer ☐ **[1]**

5. Which pair of quantities have the same S.I. base units?

A force, energy
B moment, momentum
C power, work done
D work done, moment

Your answer ☐ **[1]**

6. Two forces act in the plane of this paper.
The magnitude and direction of the two forces are shown below.



The two forces are added together.

Which diagram shows the correct resultant?

A

B

C

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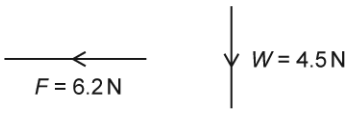
D

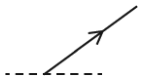
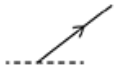
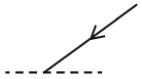
Your answer ☐ **[1]**

7. An object is in equilibrium under the action of three coplanar forces *F*, *R* and *W*. The diagram below shows the forces *F* and *W*.



The angle between *F* and *W* is 90°.

Which row shows the correct magnitude of *R* and the approximate direction of *R*?

	Magnitude of *R* / N	**Direction of *R***
A	7.7	
B	7.7	
C	10.7	
D	10.7	

Your answer ☐ **[1]**

8(a). A toy parachute is falling through air.
The air resistance *F* acting on the parachute is given by the expression

$F = kv^2$

where *v* is the speed of the parachute and *k* is a constant.

The S.I. base units for *F* are kg m s^{-2}.

Show that the S.I. base units for *k* are kg m^{-1}.

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[2]

(b). The following items of data are collected for the parachute.

- $v = 1.20 \pm 0.12$ m s^{-1}
- $F = 4.00 \pm 0.24$ N

Calculate the absolute uncertainty in k. Write your answer to 2 significant figures.

absolute uncertainty in k = ... kg m^{-1} **[3]**

9. A long metal wire is stretched between two fixed points across a laboratory bench.
The speed v of the transverse wave on the stretched wire is given by the equation

$$v = \sqrt{\frac{T}{\mu}}$$

where T is the tension in the wire and μ is the mass per unit length of the wire.

The SI base units of v, T and μ are shown below.

$v \rightarrow$ m s^{-1} $T \rightarrow$ kg m s–2 $\mu \rightarrow$ kg m^{-1}

Show that the equation above is homogeneous.

[1]

10. A drone is travelling horizontally in a cross-wind.
The drone is travelling due north at a constant speed 9.0 m s^{-1} relative to the air.
The speed of the wind is 4.2 m s^{-1}. The direction of the wind is 130° from the north.

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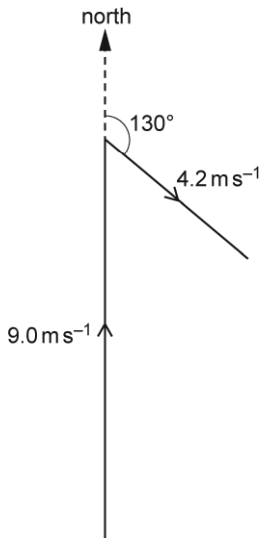
An incomplete vector diagram for determining the resultant velocity of the drone is shown below.



The vector diagram is drawn to scale.

i. Complete the vector diagram to show the resultant velocity of the drone.

[1]

ii. Determine the magnitude of the resultant velocity *v* of the drone.

v = ... m

11(a). State what is meant by a *vector quantity* and give one example.

.....

.....

..... **[1]**

(b). An object is placed on a smooth horizontal surface. Two horizontal forces act on this object.
Fig. 21.3 shows the magnitudes and the directions of these two forces.

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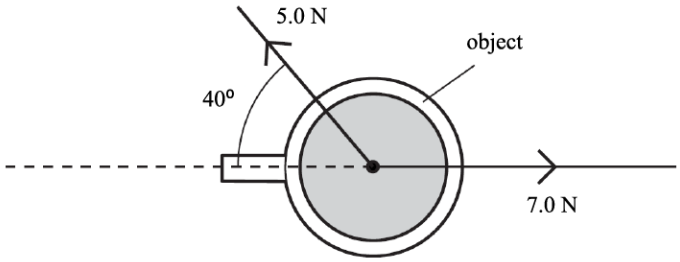

Fig. 21.3

The mass of the object is 320 g.

Calculate the magnitude of the acceleration of the object.

acceleration = …… m s^{-2} **[3]**

12. Physical quantities can be added together. Velocity and mass are examples of two different types of physical quantities. Discuss how the addition of two velocities differs from the addition of two masses.

---**[2]**

13. The unit of potential difference is the volt.

Use the equation $W = VQ$ to show that the volt may be written in base units as kg m^2 A^{-1} s^{-3}.

[3]

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14.

(a) State **one** difference between a scalar quantity and a vector quantity.

.....

..... **[1]**

(b) Fig. 1.1 shows two sets of quantities listed as 'scalars' and 'vectors' by a student.

scalars	vectors
acceleration	displacement
pressure	energy
stress	power
time	velocity
volume	weight

Fig. 1.1

(i) State the one quantity that has been incorrectly listed as a scalar.

..... **[1]**

(ii) State two quantities that have been incorrectly listed as vectors.

1. .....

2. ..... **[1]**

(iii) State two quantities listed as scalars that have the same unit. Name this unit.

1. .....

2. .....

unit: ..... **[2]**

(c) Circle the correct value for the prefix tera (T) in the list below.

10^6 10^9 10^{12} 10^{15} **[1]**

(d) Rearrange the following prefixes in the order of smallest to largest.

μ c p k

..... **[1]**

[Total: 7]

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15.

(a) Complete the table of Fig. 1.1 by stating the value or name of each of the remaining three prefixes.

prefix	value
micro (μ)	10^{-6}
mega (M)	
	10^{-9}
tera (T)	

Fig. 1.1

[3]

END OF QUESTION PAPER

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Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			D	1	
			Total	**1**	
2			**B**	1	
			Total	**1**	
3			**A**	1	**<u>Examiner's Comments</u>** This question was based on understanding the term accuracy; a key concept in practical skills. The majority of the candidates got the correct answer A. The difference between the accepted value for *g* and the experimental value was greatest for A. The most popular distractor was B, where candidates took 'least accurate' to mean the value with the least percentage uncertainty. Some candidates even had the percentage uncertainties calculated for each of the options.
			Total	**1**	
4			**D**	**1**	
			Total	**1**	
5			**D**	1	
			Total	**1**	
6			**A**	1	
			Total	**1**	
7			**A**	1	
			Total	**1**	
8	a		$v \rightarrow$ m s^{-1} **or** $v^2 \rightarrow$ m^2 s^{-2} Clear algebra leading to base unit = kg m^{-1}	M1 A1	
	b		$\frac{0.12}{1.20}(\times 100)$ or $\frac{0.24}{4.00}(\times 100)$ or (k =) 2.78 (kg m^{-1}) [2 × 0.1 + 0.06] **or** 0.26 **or** 26 %	C1 C1 A1	**Allow** (k_{max} =) $\frac{4.24}{1.08^2}$ and (k_{min} =) $\frac{3.76}{1.32^2}$ **or** 3.635 and 2.158 **Allow** (range =) 1.48

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			absolute uncertainty = 0.72 (kg m^{-1})		**Note**: The answer must be given to 2 SF – as required by the question **Ignore** any value given for *k* on the answer line
			Total	**5**	
9			$\sqrt{\frac{T}{\mu}} \rightarrow \sqrt{\frac{\mathrm{kg\,m\,s^{-2}}}{\mathrm{kgm^{-1}}}}$ clearly leading to m s^{-1} Homogeneous because *v* and $(T/\mu)^{1/2}$ have same units	**M1** **A0**	**Examiner's Comments** The proof for the homogeneity required careful progression and no omission of any key step. About half of the candidates showed sufficient rigour to score the mark. Those who lost the mark, invariably did so because of the poor manipulation of m divided by m^{-1}.
			Total	**1**	
10		i	Straight line drawn from the bottom of the 9.0 m s^{-1} vector to the end of the 4.2 m s^{-1} vector	B1	**Ignore** incorrect / omitted direction of resultant vector **Ignore** any other additional lines drawn
		ii	$v^2 = 9.0^2 + 4.2^2 - 2 \times 9.0 \times 4.2 \times \cos 50°$ $v = 7.1$ (m s^{-1}) **OR** length of resultant vector line measured **and** some calculations $v = 7.1$ (m s^{-1})	C1 A1 C1 A1	**Allow** other correct variants of this method **Note** answer to 3 SF is 7.07 **Allow** length of resultant vector in the range 5.4 − 5.6 cm **Allow** ± 0.20 (m s^{-1})
			Total	**3**	
11	a		A quantity that has both direction and magnitude. Correct example given, e.g. velocity.	B1	**Note**: The B1 mark is for a correct statement and a correct example.
	b		resultant force = $(7.0^2 + 5.0^2 - 2 \times 7.0 \times 5.0 \times \cos 40)^{1/2}$	C1	**Allow**: resultant force = $[(7.0 - 5.0 \times \cos 40)^2 + (5.0 \times \sin 40)^2]^{1/2}$
			resultant force = 4.51 (N)	C1	**Allow** full marks for a correct scale drawing to determine the resultant force; resultant force = 4.5 ± 0.1 N
			acceleration = 4.51 / 0.320 = 14 (m s^{-2})	A1	**Allow** full marks for resolving into horizontal and vertical components and combining correctly.

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			Total	**4**	
12			Mass is a scalar (quantity) and velocity is a vector (quantity). (Addition of) velocity depends on direction / sign / vector triangle / resolving (ORA)	B1 B1	**Allow** 'Velocity can be cancelled out' **Examiner's Comments** Candidates answered this opening question extremely well, with the majority gaining two marks. A variety of answers were accepted. Most candidates knew that the direction of velocities had to be considered when adding vectors. Candidates who identified mass as a scalar and velocity as a vector and then defined these two quantities were awarded full marks.
			Total	**2**	
13			(1 C =) (1) A s (1 J =) (1) kg m s^{-2} × m or (1) N = (1) kg m s^{-2} $V = \frac{\text{kg ms}^{-2} \times \text{m}}{\text{As}} = \frac{\text{kgm}^2\text{s}^{-2}}{\text{As}}$ $\text{kg m}^2\text{A}^{-1}\text{s}^{-3}$	C1 C1 M1 A0	**Allow** alternative methods **Note** this mark is for clear substitution and working **Examiner's Comments** Some candidates were not clear on what was meant by base units. Most realised that the quantity of electric charge is measured in As. Weaker candidates had difficulty deriving work done.
			Total	**3**	

14.

Question			Answer	Marks	Guidance
2	(a)		A vector quantity has direction / scalar quantity does not have direction	B1	**Not**: 'Scalar only has magnitude' because there is no mention of direction
	(b)	(	acceleration	B1	
		(ii)	power and energy	B1	
		(iii)	stress and pressure unit: pascal / Pa / N m^{-2} / kg m^{-1} s^{-2}	M1 A1	 **Note**: The A1 mark can only be scored if M1 is awarded
	(c)		10^{12}	B1	
	(d)		*p μ c k*	B1	
			Total	**7**	

15.

a	10^6 nano (n) 10^{12}	B1 B1 B1	**Allow**: 1000 000 **Allow**: nano / n / nano (N) as BOD **Allow**: 1000 000 000 000

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